



DRAFTLY

Grounded legal retrieval and a lawyer-controlled conveyancing matter workflow for Sri Lanka

GROUP 06 · UNIVERSITY OF MORATUWA · DCS&E

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Supervisor: Dr. Nisansa de Silva



A conveyancing matter has two information bottlenecks

01

LEGAL INFORMATION RETRIEVAL

One scenario can require several provisions. A missing section leaves the answer incomplete.

02

MATTER DOCUMENTS

Key facts are buried in PDFs and scans. Lawyers must read and reconcile them.

DOCUMENT REVIEW + A MEASURED SEARCH FOR COMPLETE AUTHORITY



Practitioners shaped the corpus and the matter workflow



ANURA DHANARATNA

Lawyer · Notary · Law College lecturer

3 IN-PERSON CONSULTATIONS

Notarial curriculum

→ Topic map and statutory coverage

Sample matter documents

→ Document-reading trials

Practitioner feedback

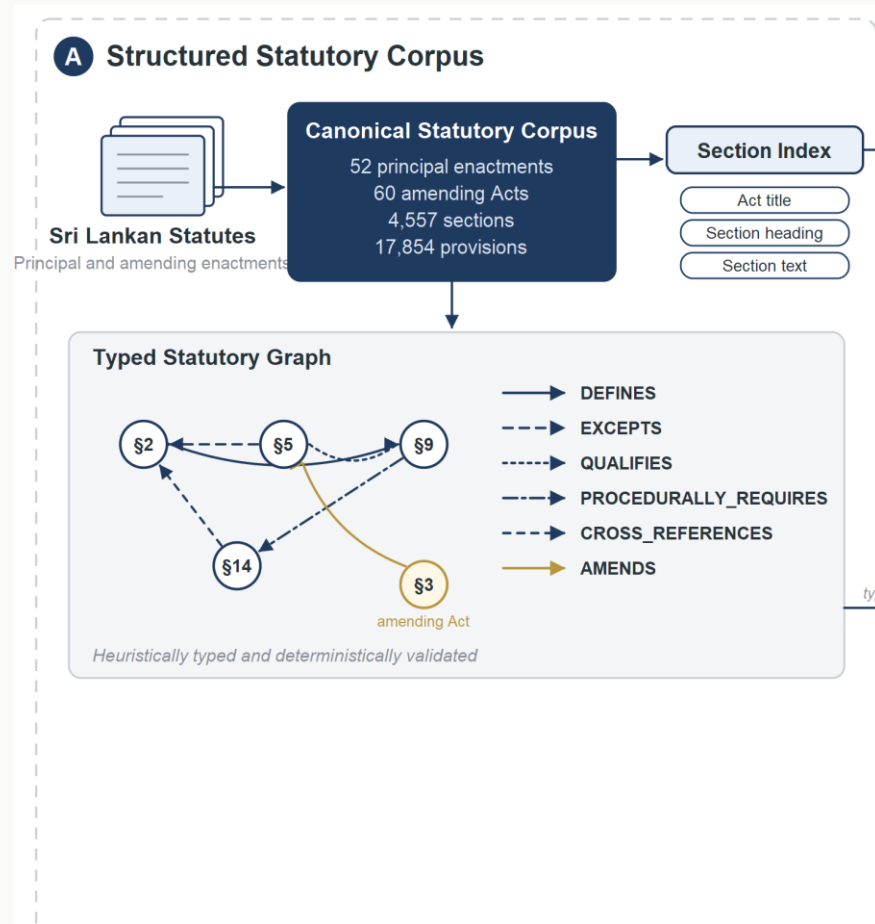
→ Evidence review and lawyer decision gates

ADDITIONAL STAKEHOLD
CONVERSATIONS





The notarial syllabus became a structured statutory search corpus



INTERNAL PARSED RESEARCH CORPUS

52 + 60

principal enactments + amending Acts

4,557

sections · 17,854 provisions

6,346

typed statutory edges



Public Act source-link index · parsed text stays internal



The case-law collection adds judgments, with provisional rule extraction

3,703

reported NLR / SLR conveyancing judgments

5,474

official Supreme Court / Court of Appeal judgment documents

3,521 / 3,703

reported cases yielded verbatim headnote rules

PROVISIONAL RULES · LAWYER VALIDATION PENDING



A separate retriever finds analogous conveyancing cases

5,121 CONVEYANCING-FLAGGED CASES / 9,177 COLLECTED



Optional dense embeddings
inactive in the recorded pilot

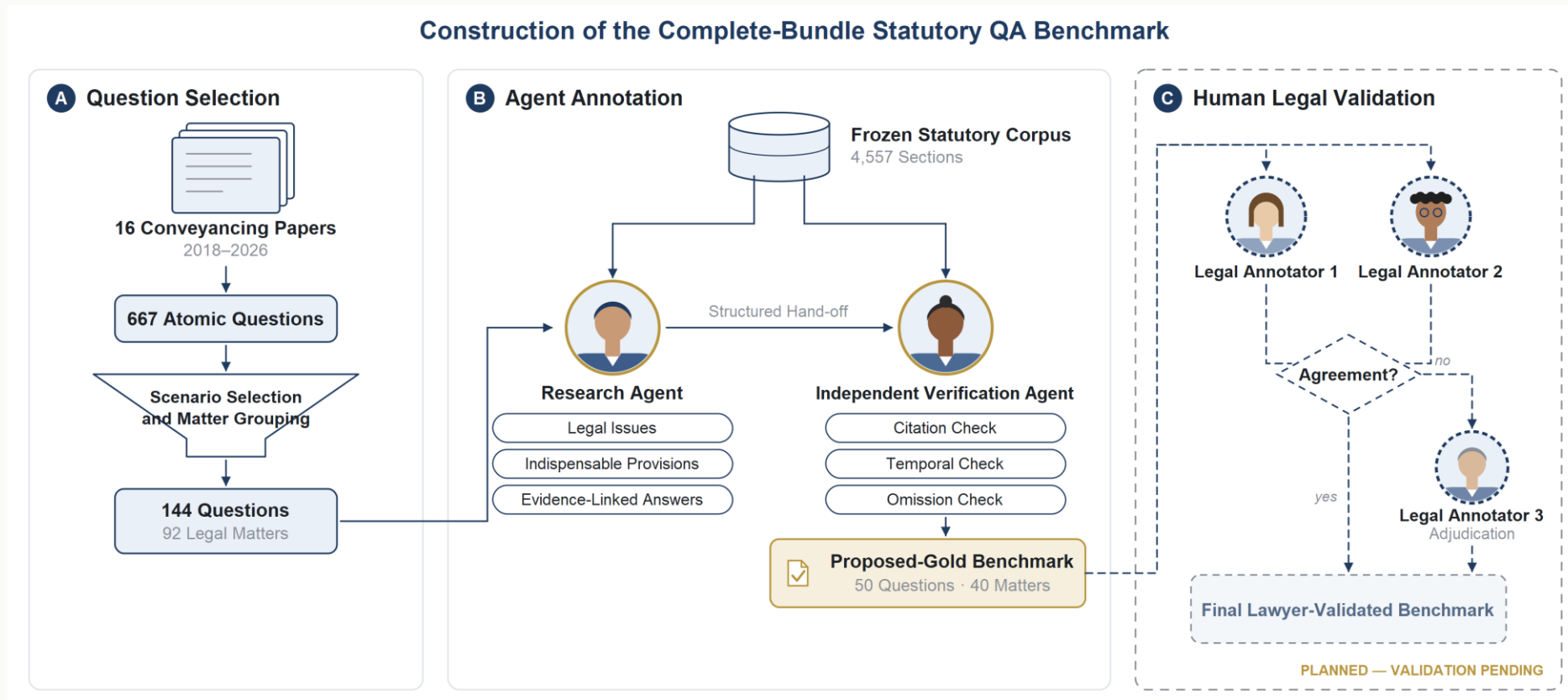
Lexical / graph corroboration
or explicit no-similar-case state

14 / 20 → 10 / 20

proxy-correct initial → later corpus snapshot; no lawyer labels



Law College scenarios became a provisional statutory QA benchmark



16 EXAM PAPERS → 667 ATOMIC QUESTIONS → 50 SCORED / 40 MATTERS



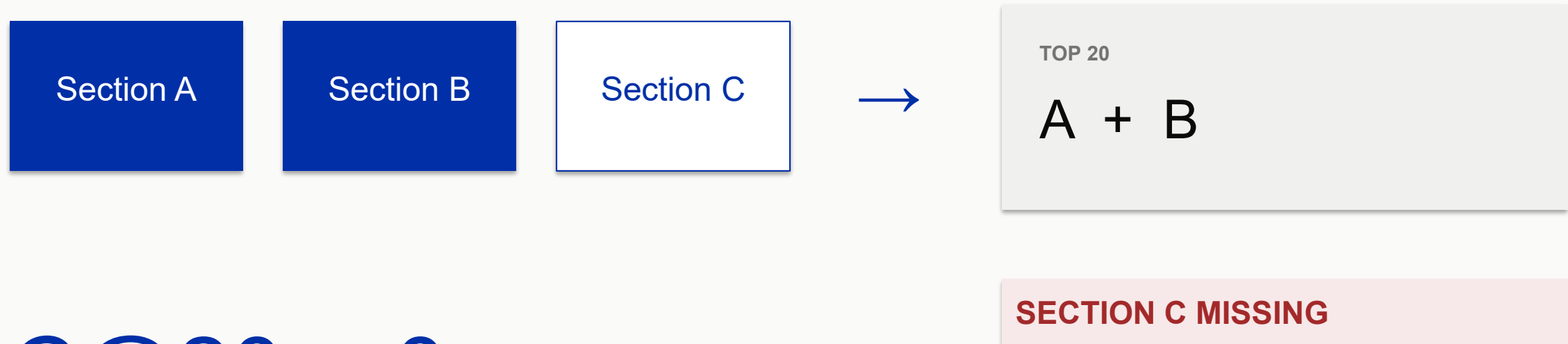
Provisional release

40 test questions / 32 matters · attorney adjudication pending



Retrieval succeeds only when it finds every required section

ILLUSTRATIVE QUESTION: AGENT-PROPOSED REFERENCE BUNDLE

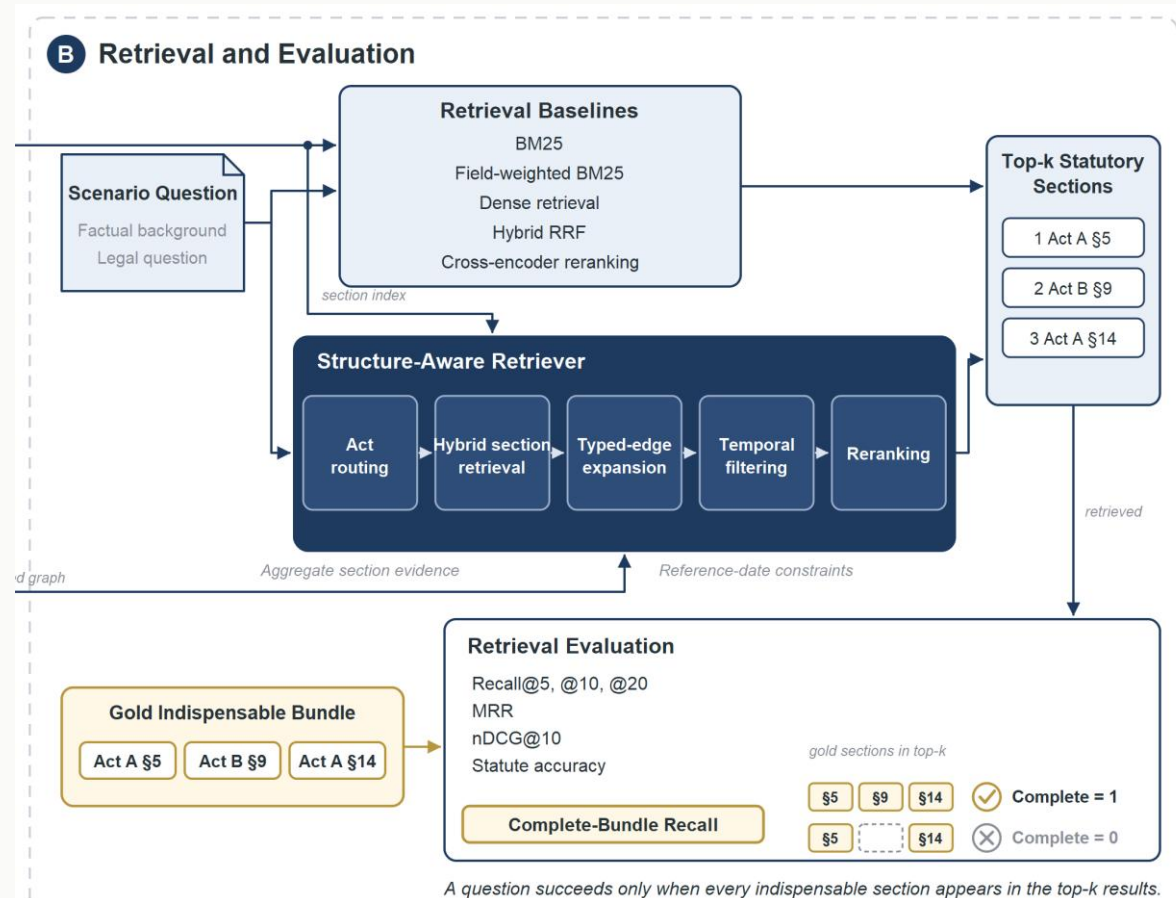


$$C@20 = 0$$

Full credit only when every proposed section appears in the first 20 results.



Seven offline retrieval systems used the same benchmark



LEXICAL

BM25 · field-weighted BM25

SEMANTIC / FUSION

Dense · hybrid · reranker

STRUCTURE

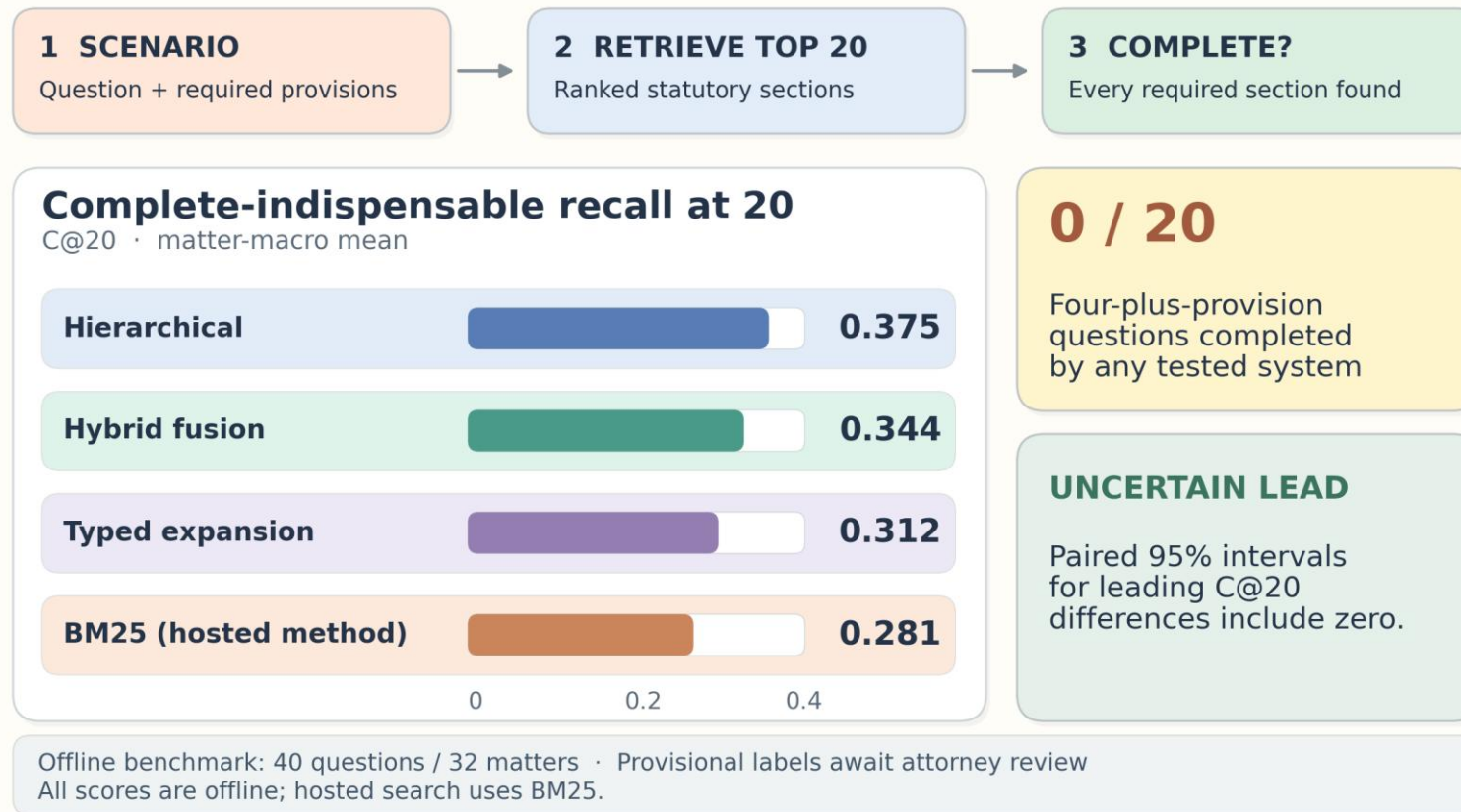
Hierarchical · typed expansion

OFFLINE RESEARCH · PROVISIONAL LABELS

Seven matched configurations · one frozen test split



Complete recall peaked at 0.375, while hard bundles remained unsolved



0 / 20

questions needing ≥4 provisions completed

0 / 20

Four-plus-provision
questions completed
by any tested system

UNCERTAIN LEAD

Paired 95% intervals
for leading C@20
differences include zero.

2 / 114

missed provisions one explicit edge away

Leading paired uncertainty intervals
include zero

OFFLINE · PROVISIONAL LABELS · 40 TEST QUESTIONS / 32 MATTERS



The accepted paper reports the complete-bundle retrieval result

Does Statutory Structure Help? Retrieving Every Indispensable Provision for Sri Lankan Conveyancing Scenarios

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ACCEPTED FOR PRESENTATION

NeurIPS 2026 GlobalSouthAI workshop

- 01 Structured Sri Lankan statutory corpus
- 02 Provisional scenario QA benchmark
- 03 Seven-system retrieval evaluation

Typed-edge expansion did not demonstrate a complete-recall gain over hybrid retrieval.



Post-certificate RTA made the first document workflow tractable

FIRST SCOPE · POST-CERTIFICATE RTA TRANSACTIONS

DEFERRED

Title certificate



Cadastral / survey plan



Transaction evidence

Handwritten pattiru / folio
OCR

AT-form extraction

Historical title
reconstruction

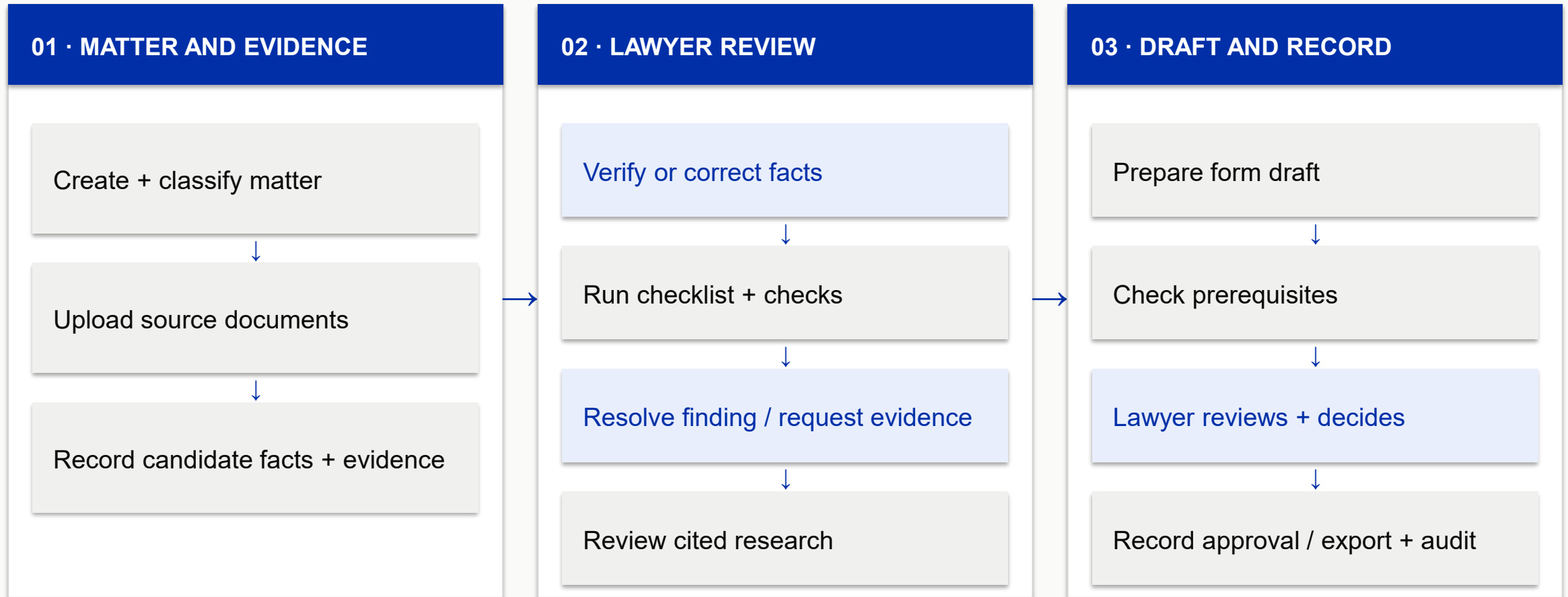
Initial title settlement

Defined requirements + prescribed forms

support reviewable checks



One matter connects intake, review, checks, and audit

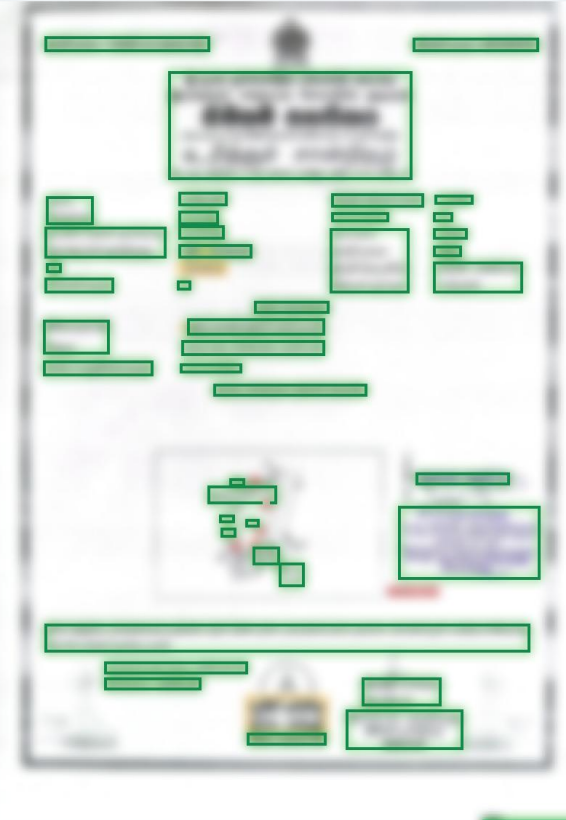


Findings return to review; a rejected draft returns to verified facts.



OCR detects text regions, but field accuracy remains unscored

REAL OCR REGION OVERLAY



Source text redacted; region boxes retained
Illustrates detection, not field accuracy

Real benchmark page · text redacted · OCR regions shown

4 / 38 / 282

matters / documents / pages in inventory

26 / 26

pilot pages returned text

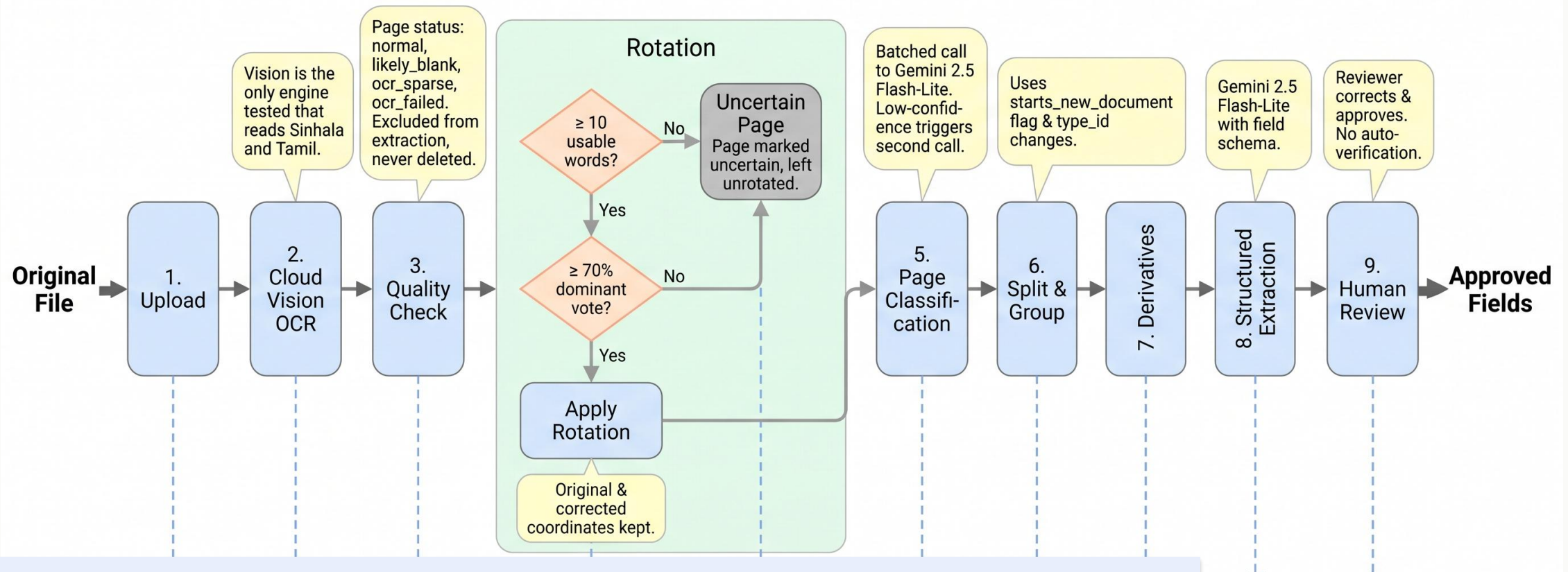
37 / 4 / 1

labelled fields / documents / matter

FIELD-LEVEL ACCURACY NOT YET SCORED



The full processing design ends at lawyer review



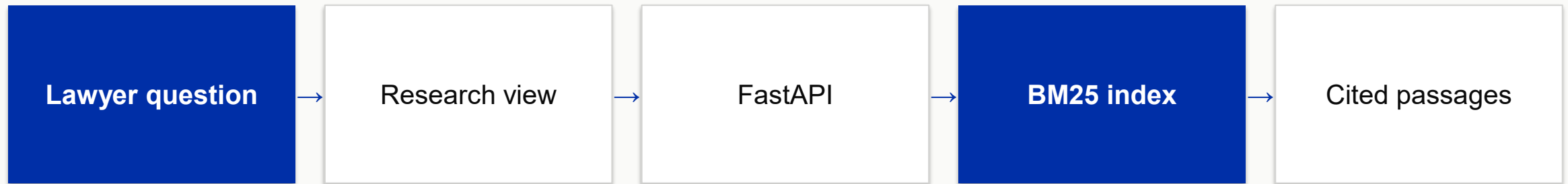
RESEARCH PIPELINE DESIGN · LATER EXTRACTION STAGES NOT HOSTED

Hosted: upload → run / stub candidates → lawyer review

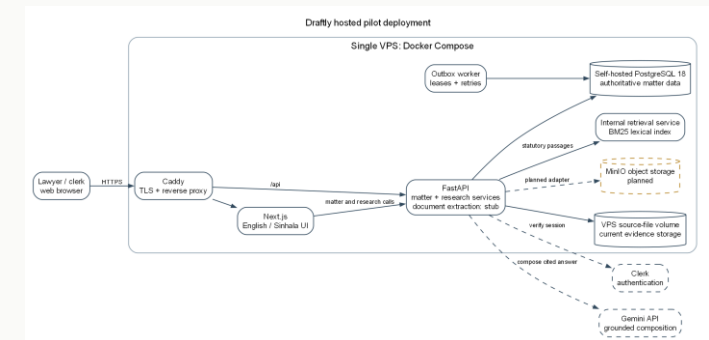


The shipped research service returns cited statutory passages

HOSTED RESEARCH PATH · BM25



Grounded response
or insufficient authority

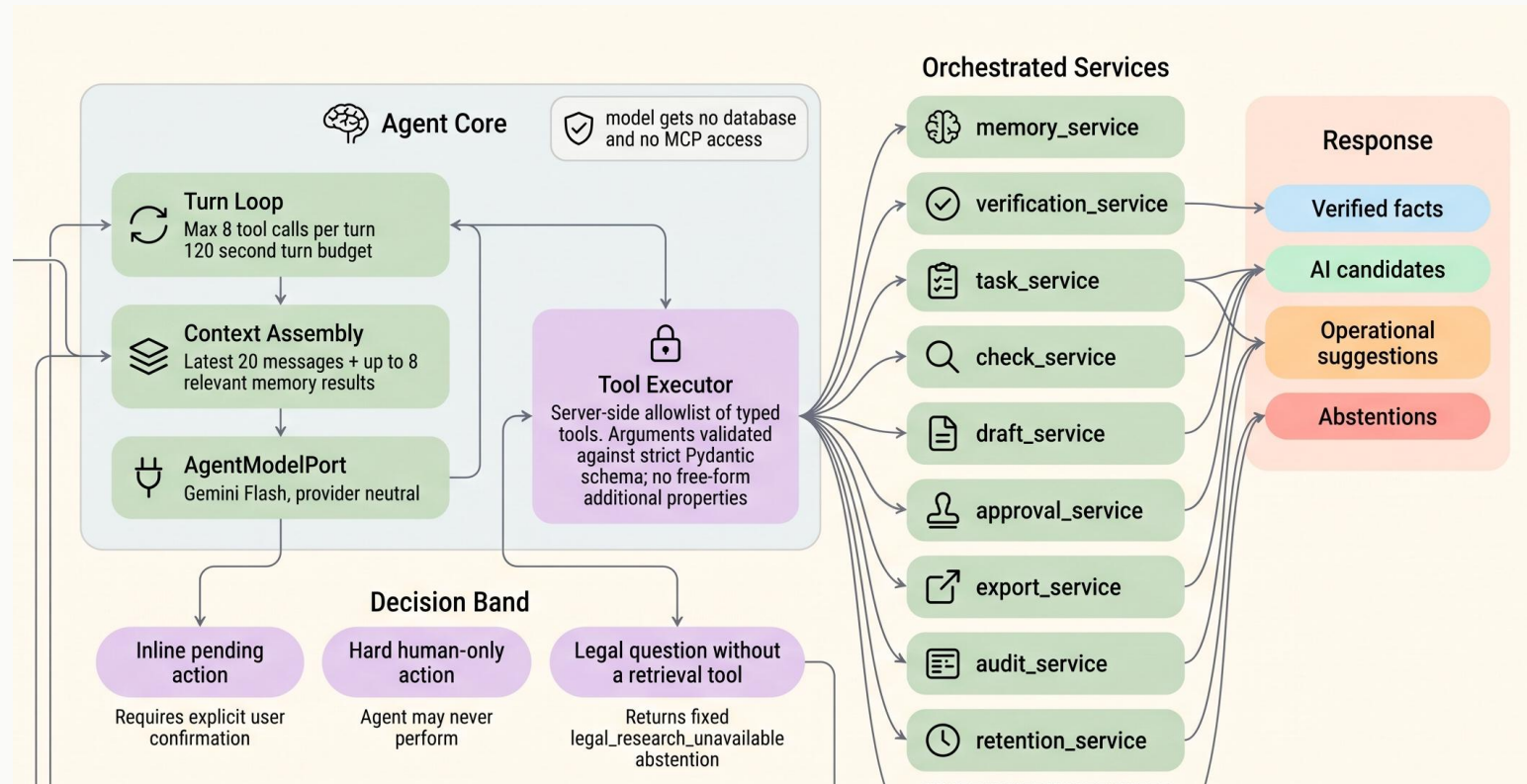


Current VPS topology

Next version in progress: embeddings + query rewriting



The assistant can propose actions; the server and lawyer control them



Matter-scoped session

Allowlisted server tools

Action → explicit confirmation

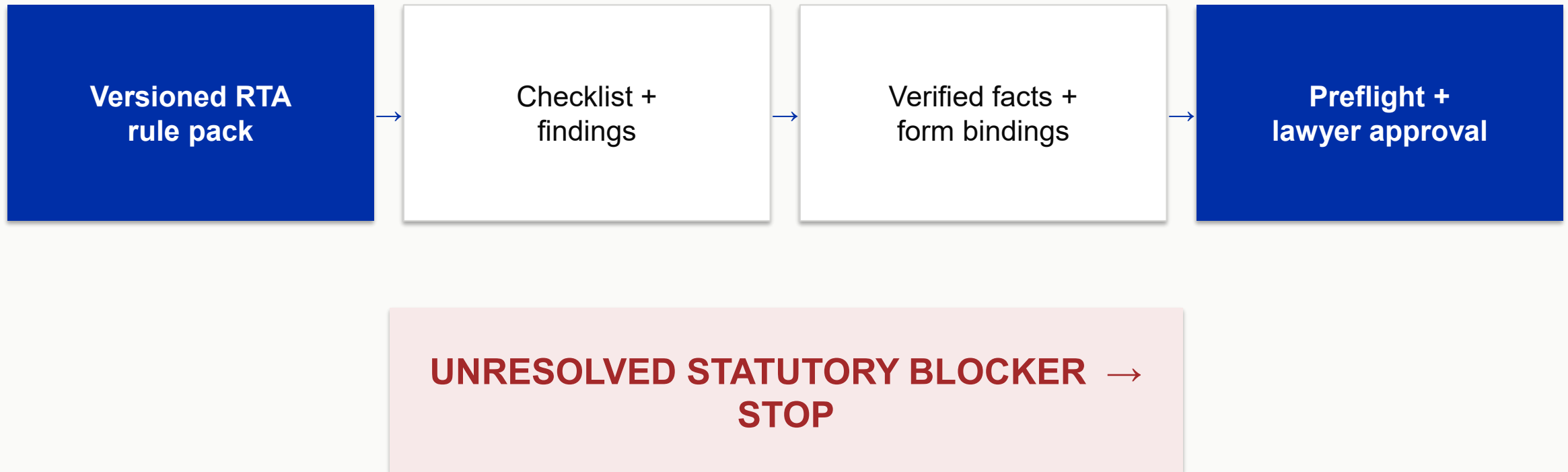
Legal question → abstention

Cited research → separate view

EARLIER DESIGN REFERENCE



The rule pack can stop an unsafe drafting path

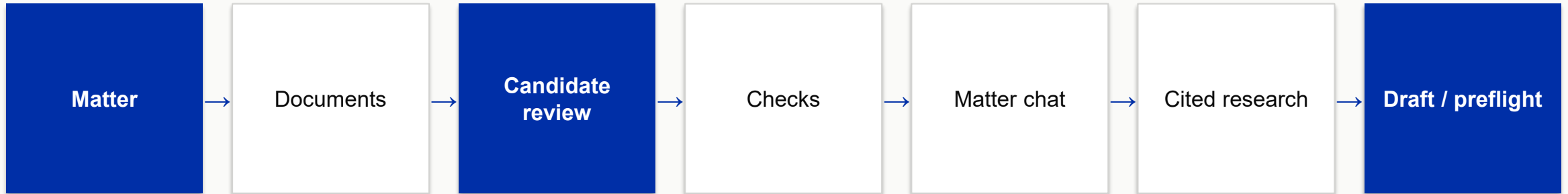


An unapproved draft cannot be exported as approved.



Inspect a synthetic RTA matter from evidence to refusal gate

LIVE PRODUCT DEMONSTRATION · SYNTHETIC RTA MATTER



Show the refusal gate and the evidence trail.

HOSTED STATUS

Stub extraction · BM25 search

Switch to the rehearsed live app or verified recording.



Five test tracks show what Draftly has measured

STATUTORY IR

7 systems · 40 questions / 32 matters

C@20 0.375 · provisional labels

SIMILAR CASES

20 queries · 14/20 → 10/20

LLM judged · dense inactive

CASE-LAW RULES

74/100 quote-grounded

58.1% proxy-usable

DOCUMENT READING

26/26 pilot pages returned text

field accuracy unscored

PLATFORM

3,906 backend + 348 frontend passed

3 browser refusal tests · one journey

WIDER BROWSER RUN: 4 PASSED · 9 FAILED · 2 SKIPPED



Three contributors connected research and the product

HIMATH DHANAPALA

- Statutory corpus + benchmark lead
- Retrieval evaluation
- High Court case-law work
- Research-paper first draft

LAHIRU DILSHAN

- Case-law collection + retrieval
- OCR research
- Document-extraction research

PRAVEEN DE SILVA

- BM25 search work
- Templates + output flow
- Lawyer-facing review interface

SHARED: INTEGRATION, TESTING, AND FINAL PROJECT REVIEW



Measured retrieval.
Versioned evidence.
Lawyer-owned decisions.

NEXT VALIDATION

Attorney labels · document fields · form review · complete browser journeys

THANK YOU